

TelemetryTrade | High-Performance Token Trading Interface



TelemetryTrade is a modern, high-performance web trading interface designed to show real-time price movement of crypto tokens. Engineered Trade Token Discovery Table and built to meet the demanding standards of low-latency traders, the project is architected with strict performance, scalability, and accessibility principles in mind.

☆☆ Features

- Avoided Next.js 15 due to React 19 conflicts; using stable Next.js 14
- Added TypeScript, ESLint, TailwindCSS, and `src/` directory structure
- Implemented App Router with `@/*` alias support
- Added delayed loader for improved UX
- Installed dependencies: `react-icons`, `framer-motion`
- Built responsive sticky navbar using `navLinks.json`
- Created interactive responsive footer using `footLinks.json`
- Integrated newsletter form with MailService + NewsletterService APIs
- Stored subscribed users in `mailUsers.json`
- `.env` added to `.gitignore` for security
- Added favicon and web manifest
- Added custom fonts: `--font-satoshi`, `--font-rubik-80s`
- Integrated shadcn/ui design system
- Implemented Skeleton Loading UI + Shimmer state
- Built responsive Hero Section
- Added Clerk secure multi-auth (email + wallets)
- Responsive Navbar with Mobile Menu

- Responsive Footer with Newsletter Section
- Installed [@tanstack/react-query](#) for data fetching and caching
- Installed [recharts](#) for visual charts
- Added [lucide-react](#) for modern icons
- Installed [tailwindcss-animate](#), [class-variance-authority](#), [clsx](#), [tailwind-merge](#)
- Built Token Discovery table with CoinGecko data, sparkline, and auto-refresh
- Added null-safe formatters and error UI in Token Table
- Used custom SVG sparkline with zero chart dependencies
- Optimized sparkline with [useMemo](#) and pure functions
- Added sorting, searching, filtering, and pagination to the Token Table
- Added currency switcher (USD/INR) with formatting
- Added table density toggle (compact/comfortable)
- Added pagination controls and footer with range display
- Added manual Refresh button and last update timestamp
- Implemented advanced sparkline with hover, tooltip, and area fill
- Added client-side Gemini chat API integration
- Built Gemini-powered Crypto Chatbot UI
- Added marketing header and disclaimers in Chat UI
- Created responsive Contact Us page with API integration and validation
- Added dark-themed global 404 page with quick navigation links
- Integrated TradingView-based CryptoChart with search and fullscreen
- Implemented 7-day sparkline modal using Recharts with stats and currency switch
- Built dark-themed TradingView candlestick widget with quick-pick and fullscreen
- Added dark-themed Terms & Conditions page with scrollspy navigation
- Enabled sticky header and active anchor highlighting for Terms page
- Ensured no metadata export in client Terms file to avoid Next.js issue
- Added links to [/privacy](#), [/disclosures](#), and legal email in Terms page
- Ensured dark/light mode support and semantic markup across the app
- Added code comments and clear changelog entries for traceability
- Modern Market Dashboard with real-time updates API
- Fully responsive down to 320px width
- Animated row deltas (green/red transitions)
- Sortable and filterable token table
- Hover effects and click-based modals
- Skeleton and shimmer loading states
- Progressive data loading
- Virtualized table rows (10k+ items with no lag)
- Lighthouse score ≥ 90 on mobile and desktop
- Connected to real trading APIs
- Optional charts powered by TradingView
- Next.js 14 with TypeScript (strict mode)
- TailwindCSS for styling
- Atomic Design folder structure
- Redux Toolkit and React Query for state and data management
- shadcn/ui + Headless UI components
- Jest + React Testing Library setup

- ESLint and Prettier configured
- SEO-friendly and accessible
- Smooth transitions with Framer Motion
- Token preview charts with Recharts
- Search and deep-link-based filters
- Full keyboard navigation support
- User-configurable columns
- Mail subscription API with MailUser storage
- Responsive sticky navbar and footer
- Custom 404 page with design consistency
- TradingView candlestick and line chart integrations
- Fully typed Contact form with backend API
- AI chatbot powered by Google Gemini
- Chat UI with optimistic updates and streaming
- Dark-themed Terms & Conditions page
- Scrollspy and anchor-based navigation
- Clerk-based multi-auth with wallet + email
- Custom SVG sparkline charts
- Skeleton UI for data fetch
- Currency switcher (USD/INR)
- Live client-side pagination with size controls
- Page number windowing with ellipsis
- Detailed price statistics in chart modals
- Toggleable table density (compact/comfortable)
- Manual and automatic refresh with status badges
- GitHub clean commits and deployment to Vercel
- Optimized folder structure under `/src`
- Web manifest, favicon, and custom Satoshi font
- Error handling panel for API failures
- Loading indicators for async components
- Reusable Button, Card, and Input components
- Utilities with `clsx` and `tailwind-merge`
- Null-safe formatting utilities
- Responsive footer with live newsletter form
- Global `.env` handling with `.gitignore`
- shadcn-based cards, modals, scroll areas
- Global provider setup for state and style
- Multi-column responsive grid layouts
- Placeholder charts with prompt to toggle to live mode
- Future scope: wallet management and deeper live APIs
- Visual regression-compatible markup, test-ready and scalable

Lighthouse Report

[Copy Link](#) [Docs](#)

Report from Nov 9, 2025, 3:00:57 PM

https://telemetrytrade.vercel.app/

Analyze

Mobile

Desktop

Discover what your real users are experiencing

No Data

Diagnose performance issues

97

Performance

91

Accessibility

96

Best Practices

100

SEO

97

Performance

Values are estimated and may vary. The performance score is calculated directly from these metrics. [See calculator.](#)

▲ 0-49

■ 50-89

● 90-100

METRICS

Expand view

● First Contentful Paint

0.5 s

● Total Blocking Time

0 ms

● Largest Contentful Paint

0.9 s

● Cumulative Layout Shift

0.002

🔥 Extras (Optional Bonus Additions)

- 🍷 Framer motion for smooth animated transitions
- 📊 Token preview charts with Recharts or TradingView widgets
- 🔄 Token state persisted via Zustand or URL params
- & Full keyboard navigation & ARIA roles for accessibility
- 🔍 Search & deep-link filters, user-configurable columns
- 🤖 Deployed on Vercel Optimised

🌐 Web Fundamentals

- Frontend:** Next.js 14 (App Router), TypeScript (strict mode), Tailwind CSS

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- **State Management:** Redux Toolkit + React Query
- **UI Components:** Radix UI / shadcn/ui / Headless UI
- **Streaming:** WebSocket (Socket.io-client mock)
- **Table:** Virtualized row rendering for 10k+ rows with no lag
- **Performance Tools:** Lighthouse, memoization, lazy-loading
- **Testing:** Jest + React Testing Library
- **Architecture:** Atomic Design (Atoms → Molecules → Organisms → Templates)
- **Styling:** Tailwind, utility-first, no inline styles
- **Documentation:** README, code comments, clean commits

🔧 Project Roadmap (24Hours Scope)

📌 PHASE 1 – Setup & Architecture

1. Initialize Next.js + TS project
2. Configure Tailwind, shadcn/ui, ESLint, Prettier
3. Add folder structure (Atomic Design)
4. Setup Redux Toolkit and React Query

📌 PHASE 2 – UI Construction

5. Build pixel-perfect token table
6. Implement tabs: "New Pairs", "Final Stretch", "Migrated"
7. Add sorting, row hover effects, modal on click
8. Build UI with Radix + Tailwind ($\leq 2px$ diff)

📌 PHASE 3 – Real-Time + Loading States

9. Mock WebSocket server to push price updates
10. Smooth animated deltas on price changes
11. Add skeleton, shimmer, progressive loading

📌 PHASE 4 – Performance & Mobile

12. Virtualized table (without layout shifts)
13. Responsive view for 320px (horizontal scroll)
14. Lighthouse score tuning ≥ 90

📌 PHASE 5 – Deliverables

15. Push code to GitHub (clean commits)
16. Deploy app to Vercel (WebSocket fix)
17. Record 1-2mi YouTube demo (public)
18. Finalize README w/ screenshots + architecture

📁 Folder Structure (Atomic Architecture)

```
TelemetryTrade/  
├── .env  
├── .eslintrc.json  
├── .gitignore  
├── components.json
```

```
├─ next-env.d.ts
├─ next.config.mjs
├─ package-lock.json
├─ package.json
├─ postcss.config.mjs
├─ README.md
├─ tailwind.config.ts
├─ tsconfig.json
├─ src/
│   └─ app/
│       ├── api/
│       │   ├── crypto-chat/
│       │   │   └─ route.ts
│       │   ├── sendEmail/
│       │   │   └─ route.js
│       │   └─ subscribe/
│       │       └─ route.js
│       ├── ai/
│       │   └─ page.tsx
│       ├── candlestick/
│       │   └─ page.tsx
│       ├── contact/
│       │   └─ page.tsx
│       ├── discovery/
│       │   └─ page.tsx
│       ├── terms/
│       │   └─ page.tsx
│       ├── fonts/
│       │   ├── Satoshi-Bold.woff2
│       │   ├── Satoshi-Medium.woff2
│       │   ├── Satoshi-Regular.woff2
│       │   └─ Satoshi-Variable.woff2
│       ├── icon0.svg
│       ├── icon1.png
│       ├── layout.tsx
│       ├── loading.tsx
│       ├── manifest.json
│       ├── not-found.tsx
│       └─ page.tsx
│   └─ components/
│       ├── ui/
│       │   ├── button.tsx
│       │   ├── button1.tsx
│       │   ├── card.tsx
│       │   ├── input.tsx
│       │   ├── light-rays.tsx
│       │   ├── scroll-area.tsx
│       │   ├── skeleton.tsx
│       │   └─ wavy-background.tsx
│       ├── ChartModal.tsx
│       ├── CryptoChatbot.tsx
│       ├── CryptoChart.tsx
│       ├── DelayedLoader.tsx
│       └─ Foot.tsx
```

```
| | | | Nav.tsx  
| | | | SlideChatSidebar.tsx  
| | | |  
| | | | data/  
| | | | | footLinks.json  
| | | | | mailUsers.json  
| | | | | navLinks.json  
| | | |  
| | | | lib/  
| | | | | utils.ts  
| | | |  
| | | | middleware.ts  
| | | | | provider.tsx  
| | | |  
| | | | public/  
| | | | | Images/  
| | | | | | Logo.png  
| | | | |  
| | | | | apple-icon.png  
| | | | | favicon.ico  
| | | | | web-app-manifest-192x192.png  
| | | | | web-app-manifest-512x512.png
```

Development Commands

```
# Install dependencies  
npm install  
  
# Run development server  
npm run dev  
  
# Run tests  
npm run test  
  
# Lint code  
npm run lint  
  
# Build for production  
npm run build
```

Contributing & Support

Clone this repository, install dependencies, and start building! For bug reports or feature suggestions, open an issue on GitHub.

License

MIT License © 2025 TelemetryTrade

Developer Notes

- Avoid Next.js 15; its forced jump to React 19 breaks half the ecosystem with dependency conflicts, while Next.js 14 stays rock-solid and production-safe.
 - or if wanna upgrade, use this overrides in config

```
"packageManager": "npm@10.5.2",  
"overrides": {  
  "react": "$react",  
  "react-dom": "$react-dom",  
  "next": "$next"},
```

- Added TypeScript, ESLint, TailwindCSS, and src/ directory structure
- Implemented App Router with alias support for @/*
- Added delayed loader for improved UX
- Installed dependencies: react-icons and framer-motion
- Built responsive sticky navbar (links via navLinks.json)
- Created interactive responsive footer (links via footLinks.json)
- Integrated newsletter form with MailService + NewsletterService APIs
- Stored subscribed user in mailUsers.json
- Added .env to .gitignore for security
- favicon with manifest
- fonts --font-satoshi --font-rubik-80s
- shadcn design system
- Skeleton Loading UI + Shimmer State
- Hero Section responsive
- Clerk secure multiauth with various wallet accounts & mail
- Responsive Navbar with Mobile Menu
- Responsive Footer with Newsletters Section
- Added @tanstack/react-query for data fetching and caching
- Installed recharts for chart visualizations
- Added lucide-react for icons
- Installed tailwindcss-animate, class-variance-authority, clsx, and tailwind-merge for enhanced styling and utility support
- Token Discovery table with CoinGecko data, null-safety, sparkline, and auto-refresh
 - WHY
 - Needed a compact crypto “discovery” table with real-time market data.
 - Existing UI had no polling, no graceful null handling, and no quick trend insight.
 - WHAT
 - Added Token Discovery page rendering top-20 tokens.
 - Pulled live data from CoinGecko [coins/markets](#) with sparkline + price % changes.
 - Auto-refresh every 60s + Next revalidate(60) for caching.
 - Graceful null/NaN handling everywhere.
 - SVG sparkline (no chart libs) + min/max hover info.
 - Inline SVG up/down badges for 1h/24h/7d % moves.
 - Skeleton rows shown on load.
 - Dark mode friendly.
 - Semantic table markup.

- LOGIC & THINKING
 - Data shaped into `Token` interface; numeric fields made `number | null`.
 - `fetchTokens()` throws if `!res.ok`; error surfaced in UI.
 - Minimal state: tokens/error/loading.
 - `useEffect`: initial load + 60s poll with cleanup.
 - Formatters return `-` when num is null/NaN. Prices `<1` use higher precision.
 - Percent fields show `+/-` sign via lightweight badge.
 - Sparkline:
 - Pure SVG; X from index, Y normalized by min/max.
 - Fallback to 2-point flat line if array missing or small.
 - `useMemo` avoids repeated geometry calc.
 - Green if last `>=` first; else red.
 - Table rows show: rank | name/symbol/image | price | 1h% | 24h% | 7d% | sparkline | market cap | volume | circ supply
 - Skeleton rows keep layout stable while loading.
- API DETAILS
 - CoinGecko endpoint:
 - GET `/api/v3/coins/markets`
 - `vs_currency=usd`
 - `order=market_cap_desc`
 - `per_page=20`
 - `page=1`
 - `sparkline=true`
 - `price_change_percentage=1h,24h,7d`
 - `locale=en`
 - Notes:
 - No API key.
 - Missing fields possible → null-safe UI.
 - Rate-limits → combined polling + revalidate to be gentle.
- EDGE CASES HANDLED
 - API failure → friendly error panel.
 - Null numeric fields → `"-"`.
 - Missing or tiny sparkline → flat fallback line.
 - `range=0` → guard to avoid div-by-zero.
 - Very small price (`<$1`) → show 6 dp.
- PERF
 - Inline SVG (no chart deps).
 - `useMemo` for sparkline geometry.
 - Minimal DOM.
- FUTURE IMPROVEMENTS
 - Sort/filter columns.
 - Currency selector + locale options.
 - Search bar.
 - Retry CTA + jittered poll.
 - Virtualize rows for `>50` assets.
- SECURITY

- No secrets; only public HTTPS API.
- No user data stored.
- CHANGELOG
 - add Token Discovery table
 - add CoinGecko fetch + polling
 - add null-safe formatters
 - add SVG sparkline
 - add loading skeleton + error UI
 - style dark/light modes
- currency switcher, sorting, filtering, density toggle, and interactive sparkline
 - Added currency selector (USD/INR) with dynamic price + compact formatting
 - Expanded API fetch to 100 assets; removed server revalidate, using no-store
 - Implemented live client-side search for name/symbol
 - Added sortable table headers for rank, name, price, pct-changes, mcap, volume, supply
 - Added row-density toggle (comfortable/compact)
 - Added manual data Refresh control + visible timestamps
 - Added status badges (asset count, last update, auto-refresh indicator)
 - Reworked sparkline to include:
 - Area fill + stroke
 - Hover crosshair + marker
 - Tooltip price display (portal)
 - Added lightweight Tooltip/HoverCard system using portals
 - Added PriceChangeBadge w/ semantic intents (success/danger/muted)
 - Improved null/NaN safety across formatting
 - Introduced memoized filtering + sorting pipeline
 - Updated UI: sticky header controls, improved dark styling
- full client-side pagination with page-size control and navigable footer
 - Added pagination state: page + pageSize (25 | 50 | 100)
 - Computed derived values: totalItems, totalPages, pageClamped, startIdx, endIdx
 - Slice filtered dataset into pageData for rendering
 - Added pagination footer with:
 - Page size selector
 - First / Prev / Next / Last buttons
 - Windowed page-number display with ellipsis for large ranges
 - Shows range badge: "Showing X–Y of N"
 - Auto-resets to page 1 when query, sort, density, currency, or pageSize changes
 - Proper disabled states + aria-label/title for navigation buttons
 - Matches table styling (dark mode, borders, hover)
- Gemini + charting deps and env config
 - Installed @google/genai, react-apexcharts, apexcharts, zod
 - Added GEMINI_API_KEY and NODE_ENV to .env.local
- Gemini-powered /api/crypto-chat with CoinGecko snapshot
 - Added POST handler with live market fetch
 - Validates env (GEMINI_API_KEY, COINGECKO_URL)
 - Reduces payload to top 20 fields for prompt
 - Generates concise trading response via gemini-2.0-flash

- Returns structured JSON output
- TelemetryAI CryptoChatbot UI
 - Added chat UI with model + user roles
 - Integrated POST /api/crypto-chat for responses
 - Added optimistic updates + scroll behavior
 - Includes loading state + Enter-to-send
- TelemetryAI chat page UI
 - Added /chat page with marketing header + disclaimer
 - Integrated CryptoChatbot component layout
 - Responsive two-column design
- Responsive Contact Us page with form validation and API integration
 - Implement two-column responsive layout (stacked on mobile)
 - Add fully typed input and textarea components
 - Connect form to /api/sendEmail endpoint using fetch
 - Include loading state and success/error message handling
- dark-themed global 404 page matching brand design
 - Created app/not-found.tsx with WavyBackground and brand typography
 - Added responsive 404 layout with quick navigation (Home + Discovery)
 - Ensured design consistency with landing page (colors, font, CTA)
- TradingView-based CryptoChart with search & fullscreen
 - Lazy-load TradingView tv.js once with cleanup
 - Manage widget lifecycle via ref; remove old widget on updates/unmount
 - Support custom symbol search (auto-append USD) and quick-pick buttons
 - Show current symbol + friendly name from POPULAR_CRYPTOS
 - Add responsive fullscreen toggle (fixed inset-0) with ESC-style control
 - Light theme, daily interval, autosize, toolbar and legend visible
 - Improve UX with input icon, focus ring, and small-screen layout
- dark-themed 7-day sparkline modal with USD/INR + stats
 - Recharts ComposedChart with dual Y-axes (USD left, INR right), lines + subtle areas
 - Memoized data shaping, min/max/avg + padded dynamic domains
 - Currency-aware tooltip/legend and compact money formatting
 - Responsive modal via portal with ESC-to-close, overlay click, and a11y roles
 - Header chips for context; configurable token name/symbol/image hook
 - Centralized USD → INR rate constant (keep in sync with app)
 - Notes: tokenImageUrl prop is wired but not rendered yet; add an for avatar if desired.
- dark-themed TradingView candlestick widget w/ search & fullscreen
 - Lazy-load tv.js once; guard if already on window.TradingView
 - Manage widget lifecycle via ref; remove before re-init; cleanup on unmount
 - Dark UI polish: neutral-900/800 surfaces, emerald accents, custom toolbar_bg
 - Set backgroundColor on widget and autosize to container
 - Symbol search with USD auto-suffix + quick-pick chips (BTC/ETH/BNB/SOL/ADA/XRP/DOT/AVAX)
 - Fullscreen toggle (fixed inset-0) with responsive chart height
 - Header copy + early-stage notice; improved focus styles
 - Notes: consider debouncing search and persisting last symbol in localStorage.
- feat(legal): add dark-themed Terms & Conditions page with sticky header, anchor nav, and scrollspy
 - New /terms page (client component) with 16 structured sections and in-page anchors

- Sticky top header and “On this page” sidebar with active section highlight (scrollspy via `getBoundingClientRect`)
- Reusable Chip and SectionCard components; glassy cards and subtle borders for dark UI
- Memoized `effectiveDate` and badges (version plus “Last updated” in footer)
- Clear legal copy: no-advice, market-data sources, fair use/rate limits, warranties, liability, and indemnity
- Links to `/privacy`, `/disclosures`, and dashboard; `mailto` for legal contact
- Fully responsive layout; improved readability with tracking, leading, and underline styles
- No metadata export in a client file (avoids Next.js “use client + metadata” error)
- Follow-ups:
 - Replace `[YOUR JURISDICTION]` and city placeholders before shipping
 - Verify `/privacy` and `/disclosures` routes exist; update contact email if needed
- Co-authored-by: — Siddhant Bali